



Sensor targets food antioxidants

A research team from Hunan City University and Xiangtan University in China has developed a sensor for detecting TBHQ, a food antioxidant used in oils and fats, addressing health concerns at high concentrations. This sensor, made



Detection of Tert-Butylhydroquinone in edible oils using an electrochemical sensor based on a Nickel-Aluminium Layered Double Hydroxide@Carbon Spheres-Derived Carbon Composite (Credit: PanuShot/Shutterstock.com)

from nickel-aluminium-layered double hydroxide (NiAl-LDH) with glucose-derived carbon spheres, enhances sensitivity and selectivity, which is key for routine food analysis. Synthesised via co-precipitation and pyrolysis at 800°C, it forms NiAl-LDH@GC-800 nanomaterials with a large surface area, boosting TBHQ adsorption. Electrochemical tests using differential pulse voltammetry (DPV) confirmed its high accuracy, with a detection range of 0.02µM-30µM and an 8.2nM limit.

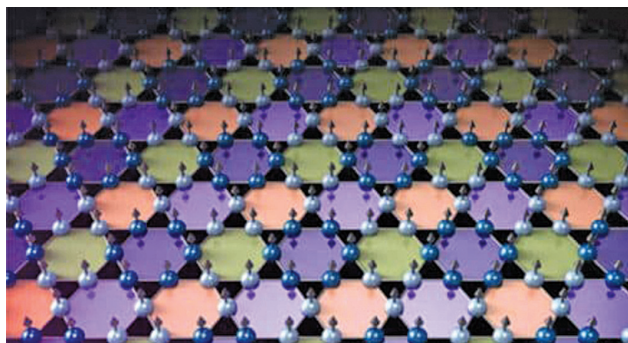
Electricity from moisture in paper

Traditional batteries are rigid and wear out quickly, while wireless power lacks range. Binghamton University researchers have developed a paper-based generator that extracts moisture from the air for power. This device uses bacterial spores to split water molecules, creating an electric charge as spores absorb moisture through paper capillaries. An added Janus paper layer, hydrophilic on one side and hydrophobic on the other, enhances moisture retention. This eco-friendly device holds promise for low-power sensors, drug delivery, and sustainable, disposable wearables.

Quantum leap in magnetism refines superconductors

Rice University physicists have uncovered key magnetic and electronic properties in kagome magnets, structures resembling basket-weaving patterns. Their research reveals that magnetism in FeSn stems from localised electron

interactions rather than mobile electrons in understanding quantum materials. Using molecular beam epitaxy and angle-resolved photoemission spectroscopy, the team produced high-quality FeSn thin films, demonstrating that



The study highlights the phenomenon of selective band renormalisation, where certain electron orbitals demonstrate stronger interactions, offering fresh insights into the intricate relationship between electron behaviour and magnetism (Credit: Princeton University and Rice University)

localised electrons govern magnetism even at high temperatures. This discovery of selective band renormalisation—where certain electron orbitals interact more strongly—advances potential applications in high-temperature superconductors, topological quantum computing, and quantum logic gates for emerging tech industries.

Data sensing with repurposed RFID tags

UC San Diego researchers have advanced passive data collection with a breakthrough in battery-free sensing. The team demonstrated that common, affordable RFID tags—



Data sensing with repurposed RFID tags (Photos from Hana Tobias)

widely used in retail—can be repurposed for real-time environmental sensing without specialised power sources. Unlike traditional battery-reliant sensors, RFID tags use analogue sensing, where environmental changes alter voltage or

current, enabling efficient, low-power data collection. Potential applications span agriculture (real-time soil moisture tracking), athletics (performance measurement in shoes), and smart parking (occupancy monitoring). This passive sensing innovation supports continuous data collection, fuelling next-generation AI and IoT advancements.